THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · NCERT-TYPE MCQ TEST SET 2

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. NCERT-type: questions in the style of the textbook examples and exercises.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. In the pattern $1, 3, 6, 10, \ldots$ what is the 9th term? 1

(a) 55 **(b)** 36 **(c)** 45 **(d)** 81

2. A protractor shows the arm of an angle at 35° on the inner scale. The reading on the outer scale for the same arm is: 1

(a) 325 **(b)** 145 **(c)** 35 **(d)** 55

3. In a supercell pyramid each number is the sum of the two numbers just below it. If the two numbers below are 13 and 18, the number above them is: 1

(a) 234 **(b)** 41 **(c)** 31 **(d)** 5

4. In the data 1, 5, 2, 2, 4, 4, 4, 4, 4, 3, 2, 2, how many values are greater than 3? 1

(a) 6 **(b)** 5 **(c)** 7 **(d)** 8

5. Using the Sieve of Eratosthenes, which is the first prime whose multiples we cross out after crossing multiples of 2? 1

(a) 1 **(b)** 4 **(c)** 5 **(d)** 3

6. How many square tiles of side 20 cm are needed to cover a floor of 400 cm by 300 cm? 1

(a) 6000 **(b)** 300 **(c)** 70 **(d)** 35

7. A pizza is cut into 12 equal slices. Riya eats 4 slices. The fraction she eats in simplest form has denominator: 1

(a) 2 **(b)** 5 **(c)** 3 **(d)** 4

8. The diagonals of a square are: 1

(a) equal and perpendicular bisectors of each other
(b) parallel
(c) equal but not perpendicular
(d) unequal

9. Which of these letters has rotational symmetry of order 2 but NO line of symmetry? 1

(a) A **(b)** O **(c)** Z **(d)** H

10. The sum of an integer and its opposite is always: 1

(a) 0 **(b)** the integer itself
(c) twice the integer **(d)** 1

11. In the pattern of cube numbers $1, 8, 27, 64, \ldots$ the term after 125 is: 1

(a) 161 **(b)** 36 **(c)** 140 **(d)** 216

12. Points lying on the same line are called: 1

(a) parallel points **(b)** collinear points **(c)** adjacent points **(d)** vertex points

13. In a supercell pyramid each number is the sum of the two numbers just below it. The bottom row is 4, 6, 3. The top number is: 1

(a) 19 **(b)** 25 **(c)** 26 **(d)** 13

14. A pictograph where one symbol = 10 shows 5 full symbols and a half symbol. The number shown is: 1

(a) 51 **(b)** 55 **(c)** 60 **(d)** 50

15. The smallest prime number that is a factor of 105 is: 1

(a) 5 **(b)** 7 **(c)** 1 **(d)** 3

16. How many centimetres are there in 3 m? 1

(a) 1300 **(b)** 290 **(c)** 300 **(d)** 310

17. 3 chapatis are shared equally among 4 children. Each child gets (of a chapati): 1

(a) $\frac{3}{4}$ **(b)** $\frac{1}{4}$ **(c)** $1\frac{1}{3}$ **(d)** $\frac{1}{2}$

18. The number of pairs of equal opposite sides in a rectangle is: 1

(a) 6 **(b)** 3 **(c)** 5 **(d)** 2

19. The smallest angle of rotation of a figure is 45°. Its order of rotational symmetry is: 1

(a) 8 **(b)** 7 **(c)** 10 **(d)** 9

20. A person has a debt of ₹40 and then earns ₹90. The balance (₹) is: 1

(a) 60 **(b)** 52 **(c)** 48 **(d)** 50

21. In the Virahanka (Fibonacci) sequence $1, 2, 3, 5, 8, \ldots$ the 8th term is: 1

(a) 55 **(b)** 21 **(c)** 47 **(d)** 34

22. What is the smaller angle between the hands of a clock at 6 o'clock? 1

(a) 270 **(b)** 150 **(c)** 180 **(d)** 210

23. Add 25 and its reverse 52. The sum is: 1

(a) 66 **(b)** 99 **(c)** 77 **(d)** 88

24. The number of times a value occurs in the data is its: 1

(a) key **(b)** frequency **(c)** scale **(d)** range

25. Which of the following is divisible by 8? 1

(a) 5130 **(b)** 5131 **(c)** 5128 **(d)** 5129

26. The area of a triangle with base 14 cm and height 6 cm is (in cm^2): 1

(a) 48 **(b)** 42 **(c)** 28 **(d)** 56

27. $\frac{2}{3} + \frac{1}{4} =$ 1

(a) $\frac{1}{6}$ **(b)** $\frac{3}{7}$ **(c)** $\frac{11}{12}$ **(d)** $\frac{5}{12}$

28. The perimeter of a rectangle with sides 11 cm and 4 cm is (cm): 1

(a) 30 **(b)** 41 **(c)** 19 **(d)** 34

29. A parallelogram (not a rectangle or rhombus) has how many lines of symmetry? 1

(a) 2 **(b)** 4 **(c)** 1 **(d)** 0

30. -9 + (-8) = 1

(a) -17 **(b)** -19 **(c)** -1 **(d)** -15

31. A complete graph has 4 points, each joined to every other point. How many lines are there? 1

(a) 16 **(b)** 10 **(c)** 12 **(d)** 6

32. A ray turns anticlockwise through 90° from the horizontal line to the right. The type of angle formed is: 1

(a) right **(b)** obtuse **(c)** acute **(d)** straight

33. Kaprekar step: for 8261, (largest arrangement) - (smallest arrangement) of its digits is: 1

(a) 7443 **(b)** 7353 **(c)** 7362 **(d)** 7344

34. In a pictograph one symbol stands for 20 units. How many symbols (counting a half symbol as half) are needed to show 70? 1

(a) 1400 **(b)** 3.5 **(c)** 3 **(d)** 4

35. Which pair is a pair of co-prime numbers? 1

(a) 14 and 25 **(b)** 9 and 15 **(c)** 16 and 24 **(d)** 14 and 21

36. A right-angled triangle has legs 5 cm and 12 cm. Its area (in cm^2) is: 1

(a) 35 **(b)** 30 **(c)** 25 **(d)** 42

37. $\frac{5}{6}$ - $\frac{1}{2}$ = 1

(a) $\frac{1}{3}$ **(b)** $\frac{1}{2}$ **(c)** 1 **(d)** $1\frac{1}{3}$

38. A circle has centre O and radius 5 cm. A point P is 8 cm from O. The point P lies: 1

(a) on the circle **(b)** inside the circle

(c) outside the circle **(d)** at the centre of the circle

39. The smallest angle through which a regular hexagon must be rotated to coincide with itself is (degrees): 1

(a) 70 **(b)** 50 **(c)** 90 **(d)** 60

40. -4 - (-9) = 1

(a) 23 **(b)** 5 **(c)** 7 **(d)** 3

41. The sum of the triangular numbers 15 and 21 is: 1

(a) 35 **(b)** 41 **(c)** 36 **(d)** 28

42. Angles $\angle AOB = 52^\circ$ and $\angle BOC = 28^\circ$ are adjacent (they share the arm OB, and lie on opposite sides of it). Then $\angle AOC$ is: 1

(a) 80 **(b)** 24 **(c)** 100 **(d)** 90

43. In the hailstone (Collatz) sequence, the number after 11 is: 1

(a) 33 **(b)** 5 **(c)** 35 **(d)** 34

44. A bar graph has scale 1 cm = 10 students. A bar showing 90 students will have height (in cm): 1

(a) 80 **(b)** 900 **(c)** 9 **(d)** 11

45. Which of these is a pair of twin primes? 1

(a) 23 and 29 **(b)** 17 and 19 **(c)** 31 and 37 **(d)** 19 and 23

46. A square has area 121 m^2. Its perimeter (in m) is: 1

(a) 33 **(b)** 66 **(c)** 44 **(d)** 55

47. Which is the greatest of $\frac{7}{9}$, $\frac{5}{7}$, $\frac{3}{4}$, $\frac{8}{11}$? 1

(a) $\frac{7}{9}$ **(b)** $\frac{8}{11}$ **(c)** $\frac{5}{7}$ **(d)** $\frac{3}{4}$

48. Two points A and B are 14 cm apart. The perpendicular bisector of AB meets AB at a point which is at what distance from A (cm)? 1

(a) 7 **(b)** 8 **(c)** 14 **(d)** 6

49. The mirror image of the letter M in a vertical mirror to its right looks like: 1

(a) the same letter **(b)** a reversed letter

(c) a different letter **(d)** an upside-down letter

50. On the number line, the integer 9 lies 9 units to the right of 0. The integer is: 1

(a) -9 **(b)** 10 **(c)** 9 **(d)** 8

END OF QUESTION PAPER **Total: 50 Marks**

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